



CY 203P: Inorganic Chemistry Laboratory

Course Code	CY 203P
Course Name	Inorganic Chemistry Laboratory
L-T-P-C	0-0-4-2
Prerequisites	None
Intended for	BS Chemical Sciences
Mutual Exclusion	None
Approval	42nd Senate (Jan 2024)

Course Contents

1. General introduction to inorganic laboratory
2. Basic concepts of quantitative analysis
3. Errors in chemical analysis data
4. Qualitative analysis: Inorganic semi micro qualitative analysis with four radicals
5. Quantitative analysis:
 - (a) Volumetric Analysis:
 - i. Acid-base titrations relevant to the neutralizing power of antacids
 - ii. Complexometric and spectroscopic estimation of metal ions
 - b) Gravimetric Analysis: • Estimation of barium/sulphate as barium sulphate
 - iii. Estimation of iron as ferric oxide etc.
6. Synthesis:
 - (a) Preparation of potash alum from scrap aluminum
 - (b) Preparation of hexamine Ni(II) chloride
 - (c) Preparation of tetramine Cu(II) sulphate

Text Books

1. A.J. Elias, **A Collection of Interesting General Chemistry Experiments**, Revised edition, Universities Press Pvt. Ltd., 2007.

References

1. **Vogel's Textbook of Quantitative Chemical Analysis**, 5th Edition, Orient Longman, 1989.
2. **Vogel's Textbook of Macro and Semimicro Qualitative Inorganic Analysis**, 5th Edition, Orient Longman, 1982.
3. Robert J. Angelici, **Synthesis and Technique in Inorganic Chemistry**, 2nd Edition, University Science Books, 1991.
4. Lab Manual and Instrument Manuals